

AI Invoice Processing Agent

AI-powered business process automation using
n8n, Google Gemini & Google Workspace

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Business Problem

Invoice processing is repetitive, manual and error-prone

Manual steps

- Receive invoices via email
- Open and read documents
- Extract data manually
- Validate amounts and fields
- Check against Purchase Orders
- Detect duplicates
- Approve / review / reject
- Maintain records and notify

Key challenges

- Manual data-entry effort
- Extraction and validation errors
- Duplicate invoice processing
- PO and invoice mismatches
- Delayed exception handling
- No centralized processing records

Project Objective

Build an agentic AI invoice workflow that receives invoice PDFs via Gmail, extracts structured data with an LLM, validates it, detects duplicates, matches POs, applies deterministic business rules, logs results and escalates exceptions to a human.

Automate invoice intake

Extract structured data

Validate invoice data

Detect duplicates

Match with POs

Apply business rules

Maintain processing log

Notify users

Human-in-the-loop escalation

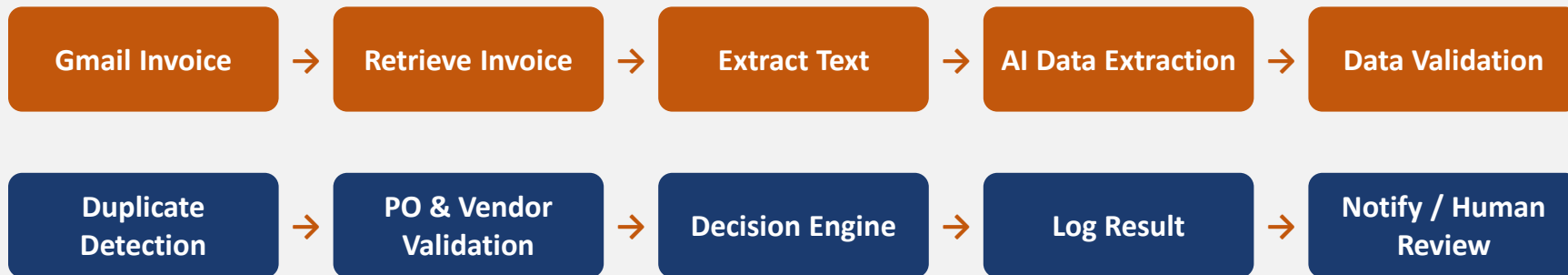
Existing Manual Process



Limitations

- Multiple manual steps and repetitive work
- Higher possibility of human error; time-consuming validation
- Manual exception identification
- Inconsistent processing records

Proposed Automated Solution



Approved / normal → Processing notification | Exception → Human review alert

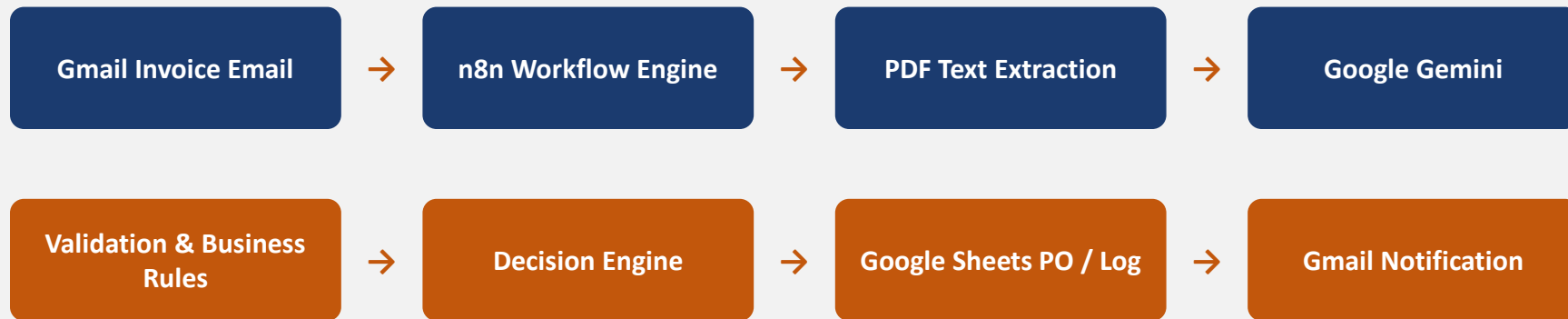
Core principle: AI handles unstructured extraction; deterministic business rules handle validation, matching and the final decision.

Technology Stack

Workflow automation + LLM + business rules + human-in-the-loop

Technology	Purpose
n8n Cloud	Workflow orchestration and automation
Gmail	Invoice email trigger and notifications
Google Drive	Storage of original invoice PDFs
Google Gemini	AI-based invoice information extraction
Structured Output Parser	Converts AI output into structured JSON
JavaScript (n8n Code)	Validation, duplicate detection, decision logic
Google Sheets	PO Master and Invoice Log
Human-in-the-Loop	Handles exceptions needing manual review

System Architecture



- Google Drive keeps the original invoice PDF
- Google Sheets holds the PO Master and Invoice_Log
- Exceptions go to a human reviewer instead of auto-approval

Final Implemented n8n Workflow

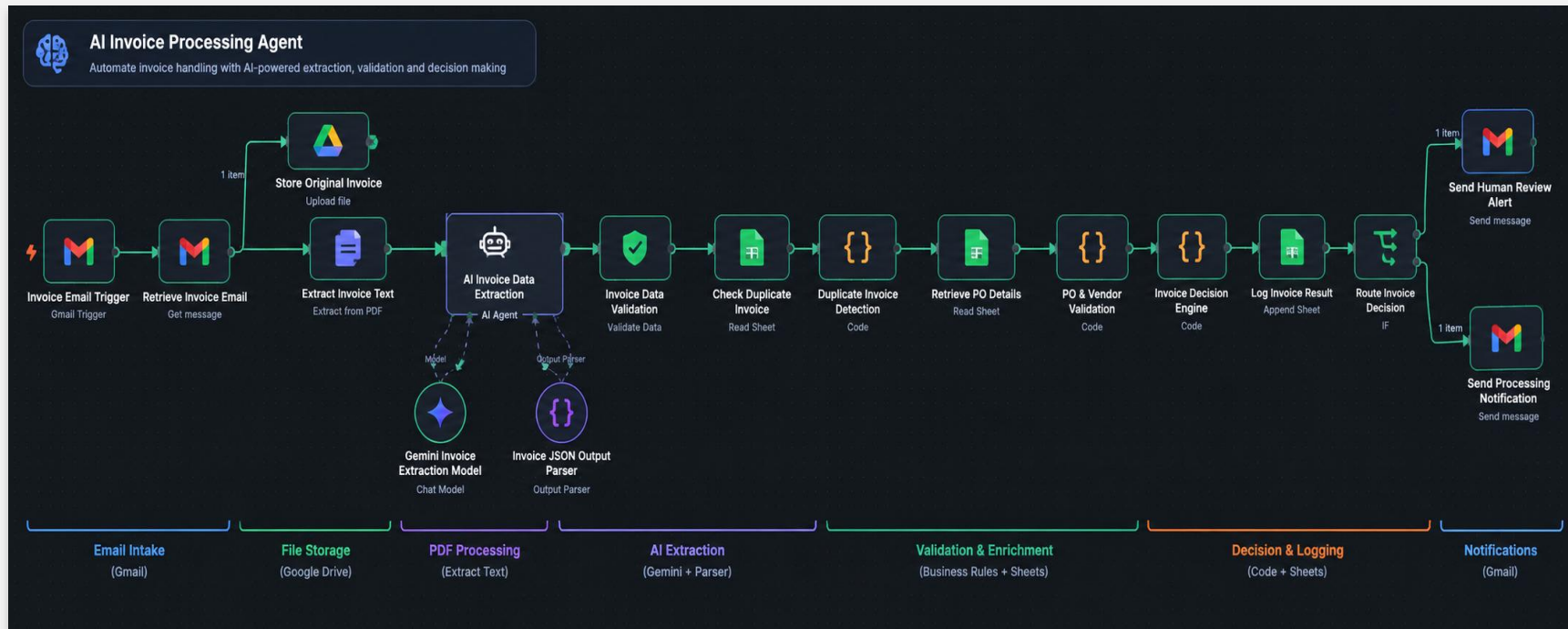


Fig. Final workflow with supporting Google Drive, Gemini and Output Parser components

- Trigger → retrieve → extract text → Gemini → validate → duplicate check → PO check → decision → log → route

AI-Powered Invoice Extraction

Gemini converts unstructured invoice text into structured JSON

Extracted fields

- Invoice number, vendor name
- Invoice date, due date
- PO number
- Subtotal, tax, total amount
- Currency

```
{
  "invoice_number": "INV-1008",
  "vendor_name": "ABC Supplies Pvt Ltd",
  "invoice_date": "2026-09-28",
  "subtotal": 40000, "tax": 5000,
  "total_amount": 50000,
  "currency": "INR",
  "po_number": "PO-501"
}
```

The LLM extracts information; it does not make the approval decision.

```
output
T invoice_number INV-1008
T vendor_name ABC Supplies Pvt Ltd
T invoice_date 2026-09-28
T due_date
# subtotal 40000
# tax 5000
# total_amount 50000
T currency INR
T po_number PO-501
```

Gemini structured output

Invoice Data Validation

Deterministic rules run after AI extraction

- Invoice number, vendor, invoice date exist
- PO number exists
- Subtotal, tax, total exist
- Subtotal + Tax = Total Amount

Example: $40,000 + 5,000 = 45,000$, but invoice total = 50,000 → INVALID → HUMAN REVIEW

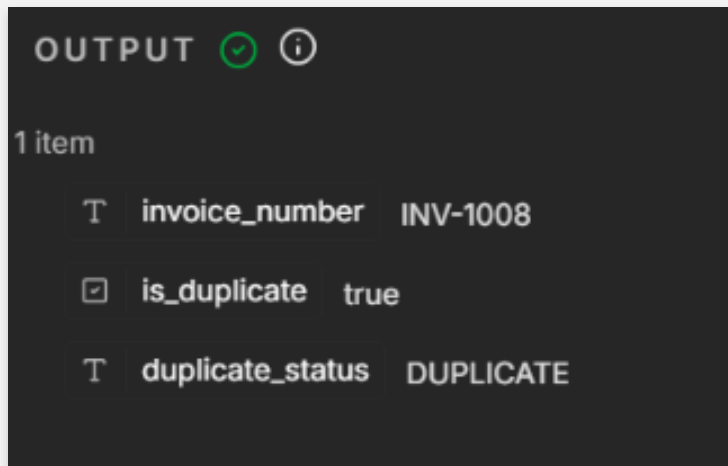
OUTPUT			Schema	Table	JSON
1 item					
T	invoice_number	INV-1008			
T	vendor_name	ABC Supplies Pvt Ltd			
T	invoice_date	2026-09-28			
T	due_date				
T	po_number	PO-501			
#	subtotal	40000			
#	tax	5000			
#	total_amount	50000			
T	currency	INR			
T	validation_status	INVALID			
T	validation_issues	Amount mismatch: subtotal + tax = 45000, but invoice total = 50000			

Validation output (INV-1008)

Duplicate Detection & PO Matching

Duplicate detection

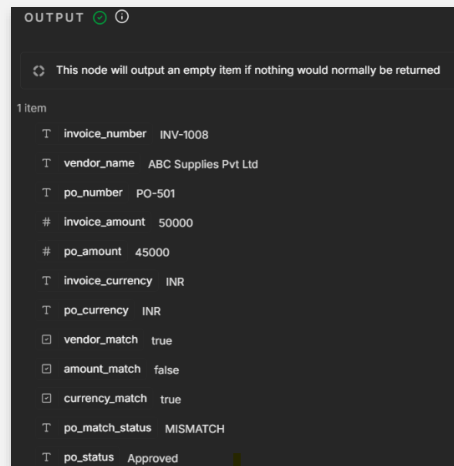
- Invoice number searched in Invoice_Log
- Already exists → **DUPLICATE** → **REJECTED**



Duplicate detection output

PO & vendor validation

- Matches PO number, vendor, amount, currency, PO status
- Any mismatch → **HUMAN REVIEW**



PO validation output (amount mismatch)

Invoice Decision Engine

Final decision comes from deterministic business rules

Rule	Decision
Duplicate invoice	REJECTED
Validation failed	HUMAN REVIEW
PO / vendor / amount / currency mismatch	HUMAN REVIEW
PO not approved	HUMAN REVIEW
All checks pass	APPROVED

The decision reason is stored for traceability.

OUTPUT	
This node will output an empty item if nothing would normally be returned	
1 item	
T	validation_status INVALID
T	validation_issues Amount mismatch: subtotal + tax = 45000, but invoice total = 50000
T	duplicate_status DUPLICATE
T	po_match_status MISMATCH
☒	vendor_match true
☒	amount_match false
☒	currency_match true
T	po_status Approved
T	decision REJECTED
T	reason Duplicate invoice detected.
T	processed_date 2026-09-28T14:44:52.558Z

Decision engine output (duplicate → REJECTED)

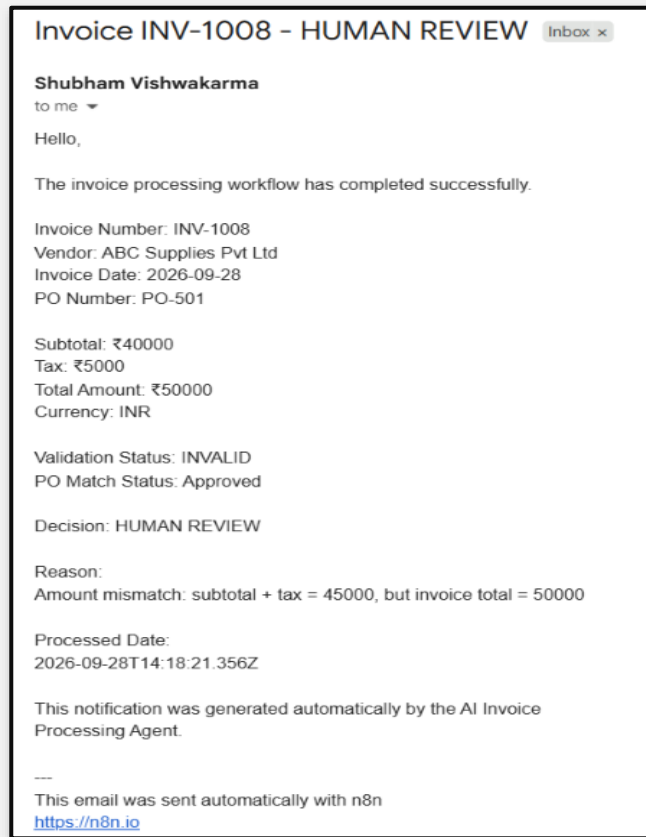
Human-in-the-Loop

Uncertain cases are reviewed, not auto-approved

A review alert is sent when

- Validation fails
- Amount, vendor or currency mismatches PO
- PO is not approved
- Other business-rule exceptions

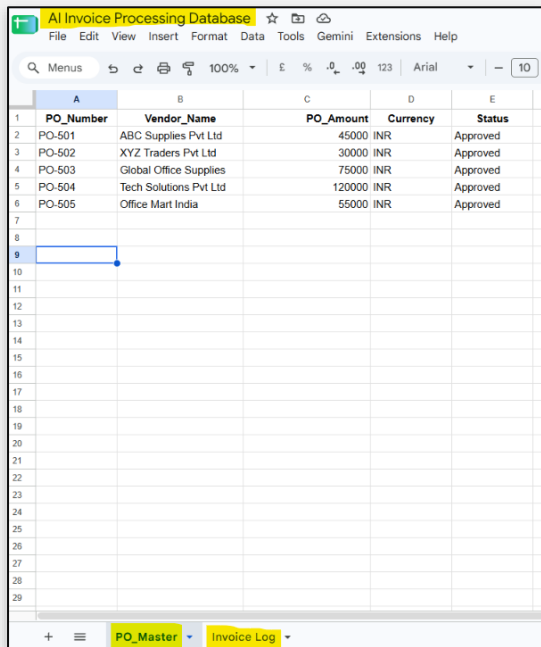
Exception → HUMAN REVIEW → logged → email alert



Human review Gmail alert

Logging, Storage & Notifications

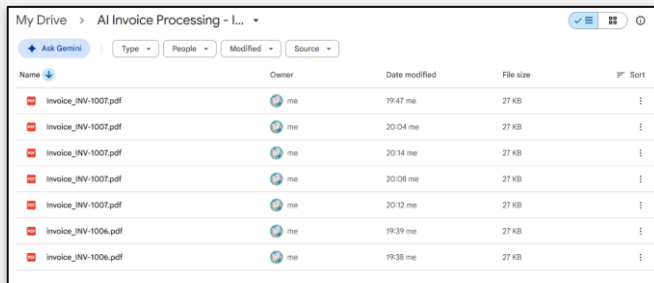
Traceability from original invoice to final outcome



The screenshot shows a Google Sheet titled "AI Invoice Processing Database". It contains a table with the following data:

	A	B	C	D	E
	PO_Number	Vendor_Name	PO_Amount	Currency	Status
1	PO-501	ABC Supplies Pvt Ltd	45000	INR	Approved
2	PO-502	XYZ Traders Pvt Ltd	30000	INR	Approved
3	PO-503	Global Office Supplies	75000	INR	Approved
4	PO-504	Tech Solutions Pvt Ltd	120000	INR	Approved
5	PO-505	Office Mart India	55000	INR	Approved

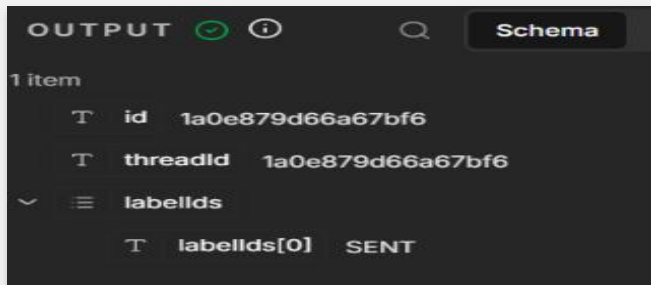
PO_Master / Invoice_Log



The screenshot shows a Google Drive interface for a folder named "AI Invoice Processing - L...". It contains a list of PDF files, all named "Invoice_INV-1007.pdf" or "Invoice_INV-1006.pdf", with file sizes of 27 KB and modification times ranging from 19:38 to 19:47.

Name	Owner	Date modified	File size	IP	Sort
Invoice_INV-1007.pdf	me	19:47 me	27 KB		
Invoice_INV-1007.pdf	me	20:04 me	27 KB		
Invoice_INV-1007.pdf	me	20:14 me	27 KB		
Invoice_INV-1007.pdf	me	20:08 me	27 KB		
Invoice_INV-1007.pdf	me	20:12 me	27 KB		
Invoice_INV-1006.pdf	me	19:39 me	27 KB		
Invoice_INV-1006.pdf	me	19:38 me	27 KB		

Original PDFs in Google Drive



The screenshot shows a terminal window with the following output:

```
OUTPUT [green checkmark] [info icon] [search icon] Schema
1 item
  T id 1a0e879d66a67bf6
  T threadId 1a0e879d66a67bf6
  labelIds
  T labelIds[0] SENT
```

Processing notification sent

- Invoice_Log stores invoice data, validation, PO match, decision, reason and date
- Original PDFs stored in Google Drive
- Human review alert or processing notification sent

Testing & Results

ID	Scenario	Expected result
TC-01	Valid invoice + matching PO	APPROVED
TC-02	Invoice amount $\neq$ PO amount	HUMAN REVIEW
TC-03	Duplicate invoice number	REJECTED
TC-04	Invoice arithmetic mismatch	HUMAN REVIEW

- **Verified:** *extraction, validation, duplicate detection, PO matching, decision logic, routing, notifications, logging*

Result: intake → AI extraction → validation → PO matching → decision → logging → notification works end to end.

Key Controls & Design Principles

Structured AI output

Parser constrains Gemini to a JSON schema

Deterministic validation

Financial checks use explicit rules

Duplicate detection

Invoice numbers checked against the log

PO matching

Vendor, amount, currency, status verified

Human-in-the-loop

Exceptions routed to manual review

Auditability

Results and reasons stored in Invoice_Log

AI extracts information; deterministic business rules control the decision.

Business Benefits

- Less repetitive manual invoice work
- Standardized validation
- Duplicate invoices detected
- PO mismatches identified
- Better exception handling
- Centralized processing records and traceability
- Automated processing notifications

Manual processing



***AI + automation +
business rules***



***Structured, controlled
processing***

Future Scope

- ERP and Vendor Master integration
 - OCR for scanned / image invoices
 - Advanced approval thresholds
 - Accounts-payable system integration
 - Multi-currency processing
- Invoice analytics dashboard
 - Human approval interface
 - Fraud / anomaly detection
 - Automated payment workflow

Conclusion

Agentic AI + workflow automation for a real business process

- Receives invoices automatically and extracts data with Gemini
- Validates data, detects duplicates, matches POs
- Applies deterministic rules and logs every result
- Sends notifications and routes exceptions to reviewers

Architecture: Gmail + n8n + Gemini + Google Drive + Google Sheets + Gmail + Human-in-the-Loop

Thank You